Query Tuning Overview

Query optimization – what the system does

Query tuning – what the programmer does

Technique summary

- Query
  - Don’t get in optimizer’s way
  - Reduce computation
  - Reduce data usage
- Design / Application
  - Denormalization (later)
  - Prefer queries over imperative code
  - Move computation out of DB
- System:
  - Improve system resources
  - Parallelize data over multiple disks
  - Don’t overuse transactions
  - Tweak system settings



Identify tuning targets

- Show query implementation
  - `EXPLAIN ANALYZE`, …
- Show query timing
  - psql: `/timing`
- Time in your client